Determining Distances to Long Period Variables in the Field of Open Star Cluster M23

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Introduction

Dr. Wilkerson and his students have been acquiring images of the field of view of open star cluster M23 for about 20 years at Luther College. These images include all stars in open cluster M23, but also other stars that appear in the field of view. These non-cluster stars have different properties than the cluster stars, including their distance from us. Some of these stars are Long Period Variables (or LPVs). Our goal is to examine a few of these LPV stars in the field of view and use the data we have on them to determine their distance from us.

Background

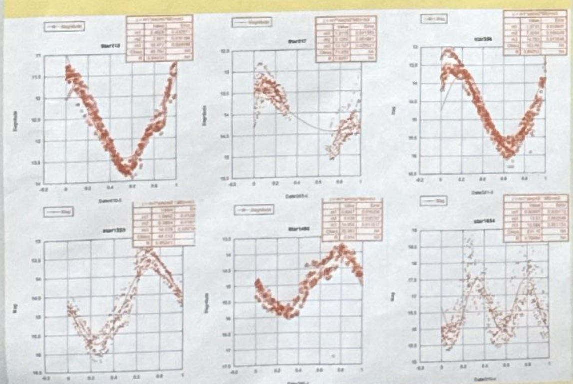
Stars that are classified as LPVs have stellar pulsation periods typically longer than 100 days. When stars pulsate, their radii expand and contract which results in varying luminosity. These pulsations can be modeled with light curves, and from these sinusoidal light curves we can estimate a period at which stars pulsate. We are concerned with Mira variable stars which are red-giants in the asymptotic giant branch (AGB) phase of their post-main sequence life. Much of the data collected in the field of M23 are multi-periodic and difficult to work with. As a result, we orient our focus to a few of the LPVs with less ambiguous light curves.

Finding the Periods of LPVs

The stars with regular light curves we observed from our field were 115, 281, 317, 356, 1223, 1495, 1654, 1702, and 1887.

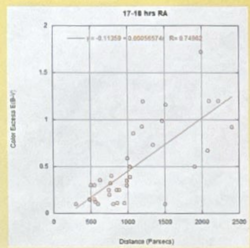
Initially we began by finding the pulsation period of the LPVs and using them to relate to luminosity. First we started looking at the light curves of each star and estimating the pulsation period in days. Next, we would divide the date (in days) by our period estimate and then translate each curve to get each pulsation overlaid with one another. From here we made minor adjustments to the period estimate until it best fit the overlaid curve. This is not the method we will use to calculate distance to our LPVs as there are not any reliable relationships between pulsation period and luminosity for Mira variable stars.

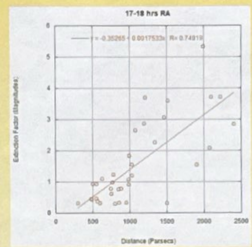
Below are the results of fitting a sinusoidal period to six of our nine stars that showed consistent pulsation periods over the many years of data collection. Here are the following periods we found for each respective LPV star:



Determining Reddening and Extinction

A common problem when determining the magnitude and color of stellar objects is that dust and gas in the interstellar medium can scatter light, which causes objects to appear dimmer and redder than they truly are. Gas and dust can be modeled as being relatively uniformly distributed in small sections of the galaxy. So we have plotted the color excess (reddening) and extinction (dimming) of 45 open star clusters that are no more than 30 minutes of RA away from M23. We can then create a fit which relates the excess color and the extinction value to the distance of a star.

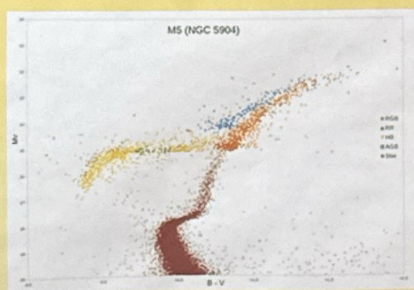




Asymptotic Giant Branch Stars

Stars in the AGB phase of their post main sequence life have burned all the hydrogen and helium from their core. They follow a predictable path on HR and color magnitude diagrams that has been well established. Knowing that our LPVs of interest should fall on the asymptotic giant branch of an HR diagram can help us back into their distances away from us. We can plot sample AGB stars on an HR diagram to use as a benchmark for where our LPV stars should also fall. Then, we plot the LPV on that same HR diagram at various distance estimates until we find the one that will put it on the asymptotic giant branch.

Below is a sample HR diagram of the globular cluster M5. The asymptotic giant branch of this cluster is in blue.



Example HR Diagram

https://en.wikipedia.org/wiki/Asymptotic_giant_branch#/media/File:M5_colour_magnitude_diagram.png

Calculating the Distances

In order to calculate the distance (in parsecs) to a celestial object, we can use the following equation that utilizes the difference in apparent magnitude *m* and absolute magnitude *M*:

$$d = 10^{((m-M)+5)/5}$$

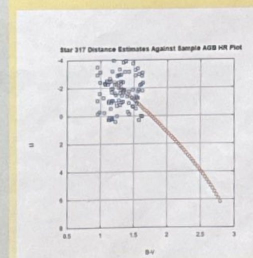
We must note that this particular equation will always overestimate distances because it does not account for the dust and gas that scatters light from stars before it reaches us. This causes the apparent magnitude we measure to be dimmer, or more positive, and therefore results in a greater distance.

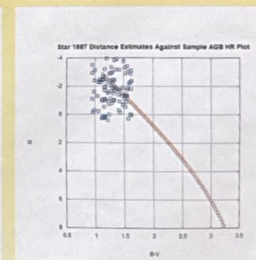
To correct for this, we use an expression that takes into account the extinction for the particular field of view (around 1 hr RA):

$$M = -5\log(d) + m + 5 - extinction$$

Our method with this equation is to guess and check various distances until we obtain an absolute magnitude, and corresponding color index, that aligns with sample AGB stars *M*s and *B-V*s on an HR diagram.

The following plots show our calculated *M* and *B-V* values for various distance estimates starting from the distance to open cluster M23 (628 parsecs) at intervals of 50 parsecs. Slight B-V translations were made to adjust for any unaccounted dust and other uncertainties in the color index. We were only able to use this method for the two stars we had B-V data for.





Conclusion

Using this incremental distance "guess and check" method, we are able to match an estimated distance to an absolute magnitude, that is corrected for extinction, for two stars of interest. The following distance estimates plotted our LPVs closest to the average sample AGB star:

- Star 317: 3278 pc
- Star 1887: 3978 pc

Our increment size was 50 pc, so we have an uncertainty of at least that much for each.

The other LPVs we were interested in didn't have B-V color indices in the given data. If we wanted to use this method to find their distance, we would need the color index data or to establish a relationship between that and the R and I color data that we do have.